Yuzhong Chen

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Education

➢ **University of Liverpool (UoL)** **Liverpool, UK**
BSc (Hons) in Computer Science (Expected) Sep. 2024 - May. 2026

- WES GPA: 3.86/4.0 | Major GPA: 4.0/4.0
- **Scholarship:** Top 50 Excellence Award (Tuition Fee Discount: £7,000 per year)
- **Fellowship:** EPSRC-funded Summer Research Assistantship, UoL, 2025
- **Test Scores:** TOEFL iBT 109 (R28, L29, S28, W24), GRE General Test 329 (V159, Q170, AW3.5, Test Center Edition)

➢ **Xi'an Jiaotong-Liverpool University (XJTLU)** **Suzhou, China**
BSc in Information and Computing Science (Expected) Sep. 2022 - Jun. 2024

- WES GPA: 3.86/4.0 | Major GPA: 4.0/4.0

Papers

➢ [1] **Y. Chen**, Y. Qu, and H. Pu, "An analysis of attention mechanisms and its variance in transformer," *Applied and Computational Engineering*, vol. 47, no. 1, pp. 164-176, Mar. 2024, doi: 10.54254/2755-2721/47/20241291. (first author)

➢ [2] **Y. Chen**, T. Tan, A. Antala, S. Salaria, T. Saha, and Q. Tang, "Sequence Matters: A Controlled Study of Modality Injection Order in Multimodal LLMs," submitted to *EMNLP 2026*, under review. (first author)

➢ [3] **Y. Chen**, B. Konev, "Modular Expert Routing System in Large Language Models: Efficiency, Specialization, and Cross-Domain Generalization," to be submitted to *ACM Multimedia 2026*. (first author)

Research Projects

➢ **MERS: Modular Expert Routing System in Large Language Models** **Liverpool, UK**
Supervisor: Prof. Boris Konev, Dean of the School of Computer Science and Informatics, UoL Sep. 2025 - Present

- Designed and implemented MERS, a modular architecture built from multiple domain-specialized expert models coordinated by a classifier-based routing mechanism, inspired by RAG and Mixture-of-Experts principles
- Initialized two baselines: unified (single LLM fine-tuned on full corpus) and modular (per-domain experts along with classifier routing) on 10 disciplinary subsets from MMLU, and benchmarked MERS for feasibility verification
- Conducted two core evaluation studies: (1) Efficiency comparison by quantifying performance, inference latency, and token-costs under unified vs. modular setups; (2) Generalization assessment by evaluating cross-domain robustness on 10 unseen MMLU subsets and characterizing the specialization-generalization trade-off with empirical evidence

➢ **Sequence Matters: A Controlled Study of Modality Injection Order in Multimodal LLMs** **Liverpool, UK**
Supervisors: Asst. Prof. Tulika Saha, UoL and *Asst. Prof. Qiyi Tang,* UoL Oct. 2024 - Jan. 2026

- Partly funded by ***EPSRC*** as a paid summer research assistantship under *School of Computer Science and Informatics*
- Proposed MOCA-MLLM, a modality order-aware controlled architecture for MLLMs, with a unified preprocessing pipeline for twelve multimodal benchmark datasets (MELD, CMU-MOSI, HateMM, RLTD, etc.), enabling order-wise injection of text, vision, and audio embeddings at predefined decoder layers for large-scale, controlled ablation studies
- Conducted extensive fine-tuning experiments across four MLLMs, six downstream tasks, and six injection sequences, revealing consistent, task-dependent modality order preferences: affective recognition tasks (sentiment, emotion) benefit from early visual/acoustic cues, whereas semantic-grounded tasks (sarcasm, humor, hate speech, deception) consistently favor early access to text, with text-middle configurations emerging as the most consistently suboptimal
- Demonstrated that carefully designed sequential modality injection consistently outperforms All-at-Input baselines, showing that increased processing depth alone does not guarantee improved performance

➢ **An Analysis of Attention Mechanisms and Its Variance in Transformer** **Beijing, China**
Supervisor: Assoc. Prof. Zengchang Qin, Beihang University Jun. - Sep. 2023

- Implemented and evaluated a suite of ML algorithms (DT, KNN, NB) and neural architectures including CNNs for license plate and traffic sign recognition, and RNNs (LSTM, BiLSTM) for sentiment and emotion classification

- Investigated Transformer variants, focusing on linearization, sparsification, multi-head self-attention and low-rank decomposition; conducted ablation studies to verify the performance improvement and highlight the trade-offs between scalability, interpretability and computational performance in Transformer-based architectures

PROFESSIONAL EXPERIENCE

➢ **Shanghai Institute of Computing Technology** **Shanghai, China**

Research & Development Center Intern Jul. - Sep. 2024

- Engineered and deployed a client-facing multimodal chatbot interface using Gradio with streaming inference and asynchronous I/O; supported text, image, file, and table inputs to demonstrate real-time reasoning and model capabilities
- Constructed a domain-specific agricultural knowledge base with LangChain and Word2Vec; optimized InternLM's RAG pipeline by integrating curated Q&A datasets and custom retrievers, improving top-k retrieval precision by 18%
- Developed a multi-turn real-time weather assistant agent, combining OpenWeatherMap API, prompt engineering and dynamic state tracking; conducted ablation studies on prompt fidelity, hallucination control and dialogue continuity

COURSEWORK PROJECTS

➢ **AI-Powered Recruitment Management System Development** Jan. - May. 2025

- Built a multi-user recruitment platform with candidate, recruiter, and admin modules; integrated OpenAI API and RAG for intelligent resume-job matching, and deployed backend via Flask and Alibaba Cloud with real-time chat functionality

➢ **Simulation and Optimization of Leader-Election Algorithms** Jan. - Mar. 2025

- Implemented distributed algorithms (LCR, HS) in Java; verified theoretical complexity bounds (O(n), O(n log n)) and reduced communication cost by up to 60% through algorithmic optimization and large-scale network simulations

➢ **Adversarial Attack and Defense in Neural Networks** Sep. - Dec. 2024

- Implemented framework (FGSM, PGD, adversarial training) on CNN baseline, ranking top 7% in robustness evaluation

➢ **Python-Based Data Visualization and Feature Analysis** Mar. - May. 2024

- Built a unified Python ML pipeline integrating feature extraction, dimensionality reduction (PCA, t-SNE, LDA), and both supervised and unsupervised learning; optimized classifier ensemble achieving top 3% performance among peers

ADDITIONAL

- **Programming Languages:** Python (3 yrs), Java (3 yrs), SQL (3 yrs), C (2 yrs), JavaScript (1 yr), HTML (1 yr)
- **Frameworks & Tools:** GitHub, Hugging Face, Jupyter, Overleaf, MATLAB, MySQL, Linux, Google Cloud Platform